

Nawaaz Khalfan

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My research is in microeconomic theory, with a focus on information acquisition, mechanism design and search. I explore the use of inspection, signalling and indexing to acquire and share information, what their use looks like in strategic environments and how these results map into applied settings.

Academic Positions

Research Fellow (Postdoctoral), Impact Labs
Monash University (2023-)
Supervisor: Simon Wilke, Advisor: Guillaume Roger

Education

Doctor of Philosophy in Economics
University of Pennsylvania (2019-2023)
Thesis: “Strategic Search and Exploration”
Supervisor: Rakesh Vohra, George Mailath, Committee: Aislinn Bohren

Master of Arts in Economics
University of Pennsylvania (2016-2019)
Thesis: “Search with Evolving Outside Options”
Readers: Rakesh Vohra, Andrew Postlewaite

Bachelor of Economics
The Australian National University (2011-2014)
Thesis: “Comparing Patent Regimes: Sequential Innovation under Incomplete Contracts”
Supervisor: Rohan Pitchford

New South Wales Higher School Certificate
Canberra Grammar School (2005-2010)
English, Mathematics, Extension Mathematics, Economics, Geography, Physics

Referees

Rakesh Vohra, Professor of Economics, George A. Weiss and Lydia Bravo Weiss University Professor, at University of Pennsylvania, rvohra@seas.upenn.edu, 215-898-6777

George Mailath, Professor of Economics, Walter H. Annenberg Professor in the Social Sciences, at University of Pennsylvania, gmailath@econ.upenn.edu, 215-898-7908

Simon Wilkie, Dean of the Faculty of Business and Economics and Head of Monash Business School, simon.wilkie@monash.edu, +61 (3) 9903 1021

Research Fields

Microeconomic Theory, Information Economics, Search and Exploration, Mechanism and Market Design, Game Theory

Research Papers

1. “Optimal Allocation with Noisy Inspection” [*working paper*]

A principal receives an unknown reward from allocating to an agent who has private information about the reward. Prior to allocating, the principal may elicit a report from the agent and inspect them at a cost, but must do so without transfers. This paper finds the principal-optimal mechanism when private information is noisy. This relates to several applied settings such as employer hiring strategies, public grant mechanisms and portfolio investment rules.

1.1. “Optimal Allocation with Noisy Verification” [*companion piece*]

This article shows that the results from “Optimal Allocation with Noisy Inspection” analogously apply to a setting where the agents have perfect information but verification is imperfect.

2. “Strategic Private Exploration” [*working paper*]

In a strategic exploration game, multiple players determine the order in which they explore unknown options with the objective of maximizing the sum of discovered rewards. Exploration is private in the sense that players cannot condition the order in which they explore on their competitor’s decisions. Equilibrium exploration procedures are determined, and losses characterized as a function of how the rewards are split when simultaneously explored. This informs us about many areas of policy design including patent and copyright regimes, R&D tournaments, and competition regulation.

3. “Sequential Information Acquisition and Optimal Search” with Rakesh Vohra [*working paper*]

A principal receives an unknown reward from allocating to one of many agents, each of whom have private information about the principal’s reward from allocating to them. The principal may elicit a report from the agents and inspect them sequentially at a cost, but must do so without transfers. This paper finds the principal-optimal mechanism when private information is noisy and the allocated good is scarce. This essentially generalises the results of Khalfan (2023) to many agents, and Ben-Porath, Dekel, Lipman (2014) to noisy types.

3.1. “Pandora’s Linear Program” [*companion piece*]

This article maps Weitzman’s canonical search problem into a linear program. This allows us to re-derive existing results concerning optimal search and extend them to problems pertaining to strategic search, information acquisition, index manipulation and robust search.

Research Projects

“Auctions and the Hold-Up Problem” with Simon Wilkie [*white paper*]

“Forecast Elicitation and Acquisition” with Guillaume Roger

“Optimal Merger Approval” with Chengsi Wang

“Manipulation Robust Search Procedures”

“Inspecting Networks”

Teaching Fields

Microeconomics, Economic Theory, Game Theory, Industrial Organization, Mechanism Design, Market Design, Information Economics

Teaching Experience

The University of Pennsylvania

As lecturer:

BDS 509: Applied Game Theory - 2022 Summer

ECON 1/LPS 601: Introductory Microeconomics - 2019 Spring, 2020 Fall, 2020 Summer

As teaching assistant:

PPE 3001: Strategic Reasoning - 2022 Fall

ECON 262: Market Design - 2021 Spring

ECON 235: Industrial Organization - 2020 Fall

ECON 1: Introductory Microeconomics - 2018 Spring, 2018 Fall

ECON 10: Introductory Business Economics - 2017 Fall, 2021 Fall

The Australian National University

As tutor and teaching assistant:

ECON 8021: Topics in Microeconomic Theory (graduate) - 2015 S2

ECON 3101: Microeconomics III – 2016 S1

ECON 3100: Economics IV Honours – 2015 S2

ECON 2141: Strategic Thinking – 2015 S2, 2016 S1

ECON 2101: Microeconomics II – 2015 S1

ECON 1101: Microeconomics I – 2014 S2, 2015 S1

ECHI 1006: The Australian Economy – 2013 S2, 2014 S1

Professional Activities

Referee for The International Economic Review

Conference presentations at the Pennsylvanian Economic Theory Conference (2022), the Australasian Economic Theory Workshop (2023, 2024), Durham Economic Theory Conference (2024)

Invited seminars at the University of Pennsylvania (2022), NYU Abu Dhabi (2023), the Australian National University (2023), Deakin University (2024)

Research Experience

Research and teaching assistant for Professor Rohan Pitchford, ANU Research School of Economics, 2015-2016

Research assistant for Professor Peter Drysdale and Dr. Shiro Armstrong, East Asia Bureau of Economic Research, ANU Crawford School of Public Policy, 2015-2016

Research assistant for Dr. Paul Chen and Dr. Martin Richardson, ANU Research School of Economics, 2016

Research and teaching assistant for Dr. Juergen Meinecke, ANU Research School of Economics, 2015

Economic Analyst, Australian Energy Regulator (AER), Australian Competition and Consumer Commission (ACCC), 2014

Scholarships and Awards

College of Business and Economics Teaching Award for Excellence in Tutoring, 2015

Economic Society Prize for Economics III, 2014

Commercial Representatives' and Agents' Association of Australia Limited Prize, 2013

Economic Society Prize for Economics II (Honours), 2012

Gold and Silver Duke of Edinburgh Award, 2009 and 2010

Canberra Grammar School Medal for Outstanding Contribution, 2010

Bill De Salis Scholarship, 2009 and 2010

Warren-Williams Award, 2009

Personal

Date of birth: September 22, 1992

Nationality: Australian

Language(s): English

Sporting Organizations: Melbourne University Athletic Club (2024-), Yarra Bend Golf Club (2023-2024), Philadelphia Runner Track Club (2022-2023), Wharton Rugby Football Club (2016-2020), Uni-Norths Owls Rugby Club (2011-2016), CGS Rugby Union (2005-2010)

Updated: September 30, 2024